RS-IML: Federated Intrinsic Mask Learning on Remote Sensing Image Interpretation

Hangyu Ye, Weiying Xie, *Senior Member, IEEE*, Xin Zhang, Yibing Lu, Yunsong Li, *Member, IEEE*, and Leyuan Fang, *Senior Member, IEEE*

***Abstract*— Interpreting remote sensing (RS) images plays a crucial role in numerous applications such as environmental monitoring, urban planning, agricultural management, and disaster assessment. Nevertheless, RS data are frequently dispersed among various organizations. Privacy concerns and data-sharing limitations make it difficult to utilize large-scale datasets within a centralized training framework. Federated learning (FL) provides a promising approach by facilitating collaborative model training across decentralized data sources, eliminating the need for data centralization. However, the application of FL in RS scenarios is challenging because the resource-limited edge nodes cannot meet the high demand for computation and memory resources to train deep learning models. Neural network lightweighting techniques have the potential to enhance model efficiency, but existing methods still present significant challenges, such as reliance on initial training of dense models during lightweighting and potential performance degradation after lightweighting. To address these challenges, we propose RS image interpretation based on intrinsic mask learning (RS-IML), a novel FL framework for RS-IML without training dense models. RS-IML comprises three key components. First, we introduce the intrinsic dimension (ID) of objective landscape that the neural network is projected onto a low-dimensional subnetwork to lightweight neural networks without training dense models. Second, we propose ID parameter averaging (IDPA) to aggregate inconsistent local intrinsic models while suppressing the adverse effects of nonintrinsic parameters between local models. Third, we fine-tune the specific parameters of local intrinsic models to mitigate global intrinsic noise for better performance. Extensive experiments demonstrate the effectiveness of our proposed RS-IML. It achieves a significant improvement in model efficiency during lightweighting compared to existing methods while obtaining superior accuracy.**

***Index Terms*— Deep learning, edge-friendly, federated learning (FL), global intrinsic noise, intrinsic dimension (ID), neural network lightweighting, remote sensing (RS).**

I. Introduction

With the rapid development of aerospace technology and sensor technology, researchers can now effortlessly gather a vast amount of high-resolution remote sensing (RS) images. The data-driven properties of deep neural networks (DNNs) makes it a popular research direction for RS data processing [1]. In recent years, DNNs provide technical support for a wide range of visual tasks in the field of RS [2], [3], [4], [5], and its powerful image processing capability is successfully applied to the interpretation of complex scenes. It is of great significance in environmental monitoring, urban planning, agricultural management, disaster assessment [1], [6], [7], [8], and other fields.

The explosive growth of RS data and the increasing precision requirements of RS missions lead to exponential growth in the scale of deep learning models. The high costs required for training hinder the application of RS technology in scenarios with limited physical space. More seriously, training DNN models on ground is increasingly infeasible because we cannot quickly download all the data from observation nodes.

1) Most RS satellites can only communicate with the ground several times a day when in contact with a ground station [9].
2) Limited storage and communication resources of edge devices such as drones are difficult to support data transmission tasks at the big data level like RS [10].
3) Downlink bandwidth is a major bottleneck—ground stations cost millions of dollars and are difficult to scale [11]. As more spacecraft and RS data compete for limited available bandwidth, the massive deployment of spacecraft clusters with high-resolution cameras further exacerbates the problem. In addition, downloading high-resolution RS images may not always be feasible due to regulatory restrictions and privacy concerns [12].

The alternative, i.e., training the models in a distributed fashion in space, presents both opportunities and challenges. Federated learning (FL) is a well-known distributed machine-learning technique for training intelligent models using decentralized data. In FL, collaborative training across multiple nodes can parallelize training, thereby significantly reducing the time required to train a neural network. It is promising for training large DNN models on a vast amount of RS data. However, a significant difference between RS distributed scenarios and typical in-center distributed settings [13], [14] is the requirement of deploying DNNs on edge devices such as small drone clusters and low orbit satellite platforms. They have a limited storage size [15], [16], [17] and power budget, while the ground center node is not sustainably accessible [17], [18], as show in Fig. 1. To enhance generalization, current DNNs often exhibit

Received 27 January 2025; revised 12 June 2025; accepted 7 July 2025. Date of publication 10 July 2025; date of current version 28 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62322117, Grant 62371365, Grant U24B20136, and Grant U22B2014; in part by the Fundamental Research Funds for the Central Universities; and in part by the Innovation Fund of Xidian University under Grant YJSJ25007. *(Corresponding author: Weiying Xie.)*

Hangyu Ye, Weiying Xie, Xin Zhang, Yibing Lu, and Yunsong Li are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: wagyr@stu.xidian.edu.cn; wyxie@xidian.edu.cn; xinzh@stu.xidian.edu.cn; yibinglu@stu.xidian.edu.cn; ysli@mail.xidian.edu.cn).

Leyuan Fang is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China (e-mail: fangleyuan@gmail.com).

Digital Object Identifier 10.1109/TGRS.2025.3587952

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

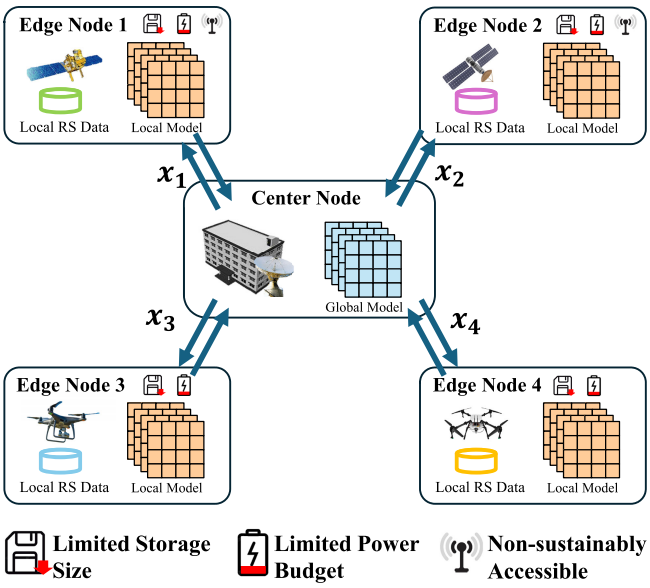


Fig. 1. Illustration of collaborative training across multiple nodes in FL for RS Tasks. RS data is typically distributed across edge nodes with limited storage size and power budget, that are nonsustainably accessible.

over-parameterization, leading to inefficiency and excessive complexity when deployed on edge devices. In addition, communication conditions are challenging due to the absence of high-end intracluster network infrastructures and reliance on wireless connections over long distances, resulting in severe restrictions on communication capabilities [19]. Therefore, reducing the training and transmission cost of FL algorithms is of paramount importance for application in RS.

To this end, much prior work has focused on integrating neural network lightweighting techniques with FL for the goal of edge-friendly frameworks. FedTiny and RigL adjustment the model structure in the center node [20], [21]. Although this approach leverages the advantages of a lightweight model, the inability of the center node to access RS data poses a risk of inadvertent deletion of crucial parameters, thereby resulting in significant degradation in the accuracy performance. FedDST and LotteryFL adjust the model structure on local nodes [22], [23]. Although these methods yield superior performance compared to the former, the pursuit of a lightweight model often entails initial training of dense models, posing challenges for edge nodes and necessitating a trade-off between model performance and efficiency in most scenarios. Therefore, it is crucial to ask *can we design a FL framework without training dense models at all while protecting accuracy performance for RS application?*

Recent research into the intrinsic dimension (ID) [24], [25], [26] of the neural network loss landscape has revealed that, owing to the over-parameterization characteristic of DNNs, models can be retrained to achieve approximately 90% of the performance of fully trained models by retaining only the essential IDs. Inspired by the theory about ID, we propose RS-IML, a novel FL framework for RS image interpretation based on intrinsic mask learning (RS-IML) to address these challenges. The key idea is to map DNNs to random low-dimensional subnetworks under the guidance of the IDs to obtain lightweight DNNs without training dense models. Allowing the nodes to participate in parameter averaging only on the ID isolates the interference between the local models in the non- ID. Parameter fine-tuning in unique IDs can effectively suppress global intrinsic noise (see 5). To assess the effectiveness and generalization of our proposed approach, we conducted comprehensive experiments across two downstream tasks in the RS domain in both independent and identically distributed (IID) and non-IID data distributions. The numerical results demonstrate that

1) The model efficiency can be enhanced by a factor of 1.7 × - 6.4 × in RS-IML compared to baseline.
2) The model efficiency of RS-IML during training is nearly twice to existing frameworks of the same type.
3) RS-IML has better accuracy performance than other comparable frameworks in most cases.

Our contributions can be summarized as follows.

1) We propose RS-IML, a novel and edge-friendly FL framework for RS that lightweight DNNs under the guidance of the IDs while prioritizing accuracy performance. To our knowledge, our algorithm is the first work that avoids training dense models during DNNs lightweighting in FL for RS applications.
2) We theoretically analyze the behavior of the model's IDs during local training and parameter aggregation. On this basis, the ID average and global intrinsic noise suppression of RS-IML are designed to prioritize accuracy performances.
3) We experimentally evaluate the effectiveness of RS-IML across two downstream tasks in the RS domain in both IID and non-IID data distributions. In addition, the sensitivity of RS-IML to important hyperparameters is tested.

II. Related Work

A. Federated Learning

FedAvg [27] is widely recognized and extensively employed in FL. It employs periodic averaging as the aggregation technique for local models of participating nodes. However, real-world FL applications encounter numerous challenges due to non-IID data distribution, node heterogeneity, and limited computational resources, memory capacity, and bandwidth [28], [29], [30].

FedProx [31] adds a proximal penalty entry to FedAvg and allows stragglers to submit partial updates to reduce the waiting latency of FL. FedGroup [32] clusters nodes into groups based on a new decomposed data-driven measure between node parameter updates. In each round of communication, each active node contributes only its local optimization results to the corresponding group model. This method can be combined with FedProx to achieve better performance. FedAvgM [33] shows that adding a momentum item to the local optimizer can consistently improve performance in FL. Inclusive FL [34] can train a large model over devices with heterogeneous capabilities. For FL with high communication efficiency, Wu et al. [35] introduce knowledge distillation technology into FL and propose FedKD, which achieves competitive results with less communication cost. CSGD [36]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

optimizes communication costs by enabling nodes to participate in global aggregation only when they possess a certain amount of information.

B. Federated Learning in RS

The dramatic increase in RS data poses considerable challenges for efficient data handling and processing, which are critical for enabling various RS applications [37], [38], [39], [40]. As the volume and velocity of data generation in RS continue to grow, processing this data has increasingly become a big data challenge [41], [42], [43]. In recent years, deep learning has become a dominant paradigm in RS image interpretation, enabling significant breakthroughs in tasks such as classification, super-resolution, and domain adaptation. Early convolutional neural network (CNN)-based approaches leveraged hierarchical feature extraction to achieve improved performance over traditional methods. However, recent advancements have introduced more powerful architectures. For instance, enhanced unmixing-inspired unsupervised network with attention-embedded degradation learning have been proposed for unsupervised hyperspectral image super-resolution [44], introducing a hybrid model-constrained loss containing a perceptual abundance term and a degradation-guided term to further eliminate the latent distortions. Transformer-based models, such as ViT combined with supervised contrastive learning, have also demonstrated strong performance in RS scene classification by better modeling long-range dependencies [45]. Furthermore, to address the challenge of cross-scene generalization, multisource collaborative domain generalization has been introduced to improve robustness in multidomain classification tasks [46]. Consequently, recent research efforts have increasingly concentrated on developing distributed and parallel processing methods for handling big data in the field of RS. Because of its capability to support distributed data processing while ensuring data privacy, FL has become increasingly crucial for RS applications.

RS-DCNN [47] is a distributed deep learning-based classification method designed for large-scale RS datasets. FedFusion [48] is a manifold-driven multimodal fusion framework including a FL module for iterative updates of network parameters across nodes via global weighted averaging. To reduce communication cost, they employed low-rank decomposition techniques. This approach enables efficient multimodal fusion while minimizing interclient communication costs. FedPM [49] enhances object extraction in deep CNNs through prototype matching. Likewise, GeoFed [50] addresses semantic segmentation in Earth observation by optimizing objective function with essential feature mining strategy.

Additionally, the combination of a deep memory connected neural network with a data-decoupled FL framework [51] supports image restoration while ensuring data privacy. An adaptive model communication scheme [52] employs deep Q-learning to optimize resource control in multiaccess edge computing environments. This scheme uses an epsilon-greedy strategy to allocate computational resources efficiently. To enhance both security and efficiency,

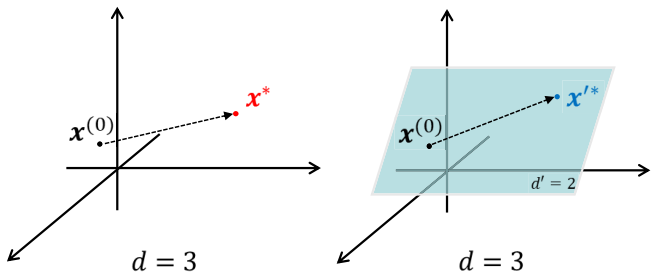


Fig. 2. *(Left)* illustration of parameter vectors for direct optimization in the $d = 3$ case. *(Right)* illustration of parameter vectors and a possible random subspace.

FLBIC-CUAV [53] integrates clustering, blockchain, and FL. It leverages beetle swarm optimization (BSO) for UAV clustering, blockchain for secure data transmission, and a residual network model within FL for cloud-based image classification. Similarly, a blockchain-empowered privacy-preserving FL (PPFL) framework [54], utilizing the CKKS cryptosystem [55], allows satellite imagery owners to collaborate globally while ensuring data privacy.

For improving classification performance across diverse data modalities, a multimodal FL framework [56] associates images from different nodes with various modalities, thereby enhancing model robustness. Additional strategies, such as feature-centric communication and pseudo-weight amalgamation, have been explored to improve the efficiency of model aggregation in FL [57]. FedDiff [58] presents a novel multimodal diffusion-based FL framework that combines dual-branch diffusion models for feature extraction with a lightweight communication module. This ensures efficient and private collaboration among nodes. These advancements highlight the growing sophistication of FL in RS, as it continues to tackle diverse challenges while maintaining data privacy and improving model performance. Moreover, FedRSCLIP [10] improves both representation learning and classification accuracy while enhancing communication efficiency, thereby tackling issues like excessive communication overhead and data diversity.

C. ID in Deep Networks

Li et al. [25] define the ID of the objective landscape to quantify and significantly reduce parameter redundancy by constraining optimization to a fixed stochastic projective space with much smaller dimensions than the original parameter space. Fig. 2 shows a simple example diagram of this redundancy reduction [25]. In the left panel, the parameter space is 3-D (i.e., $d = 3$), and the model is directly optimized from the initial point $x(0)$ toward the global optimum x^* in the full space. In the right panel, we still operate in a 3-D parameter space, but the optimization is restricted to a 2-D intrinsic subspace (i.e., $d = 2$), which is represented as a shaded plane. The parameter vector is optimized toward $x^{*'}$, the projection of the global optimum onto the intrinsic subspace. This subspace-constrained view underlies the intrinsic mask learning approach: the intrinsic mask selects a low-dimensional trainable subspace where optimization occurs, enabling parameter efficiency while preserving task performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

In recent years, the ID is often used to measure model efficiency [24], [25], [26]. It is closely related to the degrees of freedom of the parameter space [59], [60] and compressibility [61]. Janson et al. [59] investigate the degrees of freedom [62] in deep networks and the expected difference between test error and training error. LDMNet [63] explicitly penalizes the ID regularizing the network training. Cheung et al. [64] use ID to superimpose multiple models into one model. Birdal et al. [65] use the recently developed connection between fractal dimensions and topological data analysis to efficiently calculate the IDs of networks in various settings and to predict generalization errors. Gressmann et al. [66] propose random bases descent (RBD) by reoptimizing in low-dimensional random subspaces to improve their actual optimization performance. Based on relevant theories, numerous advanced machine-learning algorithms have been developed to acquire lightweight and efficient neural network models in an easy way [67], [68], [69].

D. DNNs Lightweighting in FL

For FL on edge devices, many efforts attempt to combine lightweight technologies with FL. LotteryFL [23] applies the theoretical results of the lottery hypothesis to local training of nodes to obtain lightweight multiple personality models. FedTiny [20], inspired by RigL [21], performs pruning and growth on the device, and the center node generates a new global model through sparse aggregation. FedDST [22] turns dynamic sparse training into FL, seamlessly combining sparse neural networks and FL paradigms. However, the above methods face difficulties in practical application because they require training dense models on edge devices, requiring high computing resources and storage resources. As such, they are not suitable for cross-device FL [28], where the node is a resource-constrained device.

Various techniques can be integrated into RS-IML for better application results. Although other methods, such as communication topology design [70], decentralization [71], and gradient compression [72], [73] are not considered in this article, these orthogonal techniques can definitely be applied to RS-IML.

III. Methodology

In our approach, we assume that there are N edge nodes locally storing their RS data and a center node, collaborating to solve the following optimization problem:

$$\min_{\boldsymbol{x}\in\mathbb{R}^d} \frac{1}{N}\sum_{i=1}^{N} [l_i(\boldsymbol{x}) := \mathbb{E}_{\xi_i\sim D_i}\mathcal{L}_i(\xi_i\ , \boldsymbol{x})] \tag{1}$$

where $\boldsymbol{x}$ denotes model parameters, and ξ_i denotes the local data sampled at node i that follows distribution D_i. The expectation of the loss function $\mathcal{L}_i(\xi_i\ , \boldsymbol{x})$ of node i is denoted by l_i. In the preparatory stage, single communication is employed to generate both local and global intrinsic masks. In distributed training stage, participating nodes receive the latest global intrinsic model to initialize their local intrinsic model and perform local training on the local intrinsic model with local data. Subsequently, nodes mitigate the impact of global intrinsic noise in the unique dimension of local intrinsic models. Finally, local intrinsic models are uploaded to the center node for ID parameter averaging (IDPA) to update the global intrinsic model. The complete flow of RS-IML is elaborated in Algorithm 1 and Fig. 3. The following is an introduction to the three key components of RS-IML: intrinsic mask generation (IMG), IDPA, and global intrinsic noise mitigation.

Algorithm 1 RS-IML

Require: nodes $[N]$ with local data ξ_i, communication round R, epoch $E = [E_{\text{IMG}},\ E_{\text{Local_train}},\ E_{\text{Mitigation}}]$, node sampling ratio C, learning rate η, hyperparameters $Acc_{\text{threshold}}$ and α.

Initialize: global model $\boldsymbol{x}^{(0)}$, parameter masks $\boldsymbol{\mathcal{M}_i}$

Center node broadcasts the initial global model $\boldsymbol{x}^{(0)}$ and random parameter mask $\boldsymbol{\mathcal{M}_i}$ to all nodes

$\hat{\mathcal{M}}_{\mathbf{i}} \leftarrow \text{IMG}(\ \xi_i, \boldsymbol{x}^{(0)}, \mathcal{M}_i, E_{\text{IMG}}, Acc_{\text{threshold}}\ )$ Nodes generate local intrinsic masks as shown in Algorithm 2

$\hat{\boldsymbol{\mathcal{M}}}_{\text{global}} \leftarrow \sum_{i=1}^{N} \hat{\boldsymbol{\mathcal{M}}}_i$ Center node collects all $\hat{\boldsymbol{\mathcal{M}}}_i$ and calculates the global intrinsic mask.

$\hat{\boldsymbol{x}}_{\text{global}}^{(0)} \leftarrow \boldsymbol{x}^{(0)}$

for $r = 1, 2, \ldots, R$ **do**

  $S^{(r)} \leftarrow$ Sample $N \times C$ nodes.

  $\hat{\boldsymbol{x}}_i^{(0)} \leftarrow \hat{\boldsymbol{x}}_{\text{global}}^{(r)}$ Center node broadcasts the current global intrinsic model $\hat{\boldsymbol{x}}_{\text{global}}$ as $\hat{\boldsymbol{x}}_i^{(0)}$

  for $k = 1, 2, \ldots, E_{\text{Local_train}}, node\ i \in S^{(r)}$ **do**

    $\hat{\boldsymbol{x}}_i^{(k)} \leftarrow \hat{\boldsymbol{x}}_i^{(k-1)} - \eta \nabla l_i(\hat{\boldsymbol{x}}_i^{(k-1)})$

  end for

  for $k = 1, 2, \ldots, E_{\text{Mitigation}}, node\ i \in S^{(r)}$ **do**

    $\hat{\boldsymbol{x}}_i^{\prime(k)} \leftarrow$ (Node i adds a random noise disturbance to parameters with $(a = 0, b \geq 1)$.

    $\left(\hat{x}_i^{(E_{\text{Local_train}}+k)} \leftarrow\right.$ Fine-tune the $(a = 1, b \geq 1)$ parameters with $Loss_{\text{mitigation}}$ in Equation 8.

  end for

  $\hat{\boldsymbol{x}}_{\text{global}}^{(r+1)} \leftarrow$ Center node updates the global intrinsic model according to Equation 4.

end for

A. Intrinsic Mask Generation

The ID of the network is defined by the minimal number of parameters required to represent the data manifold effectively. The mask generation process identifies this minimal set of parameters through training and reduces the model's complexity by selectively keeping the parameters that correspond to the ID, ensuring the model remains both efficient and accurate.

In the preparatory stage, the center node broadcasts the initialized global model to all edge nodes with randomly initialized sparse parameter masks. $\boldsymbol{\mathcal{M}}_i = \{m_{i,j}\}_{j=1}^{d}$ denotes the mask of node i, which is a sparse zero-one tensor corresponding to the network structure. In this article, 5% of entries are set to 1 for initialization. It effectively projects the full network onto a local subnetwork with reduced dimensions

$$\boldsymbol{\Theta}(\boldsymbol{x}, \boldsymbol{\mathcal{M}}_i) = \boldsymbol{x} \times \boldsymbol{\mathcal{M}}_i. \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

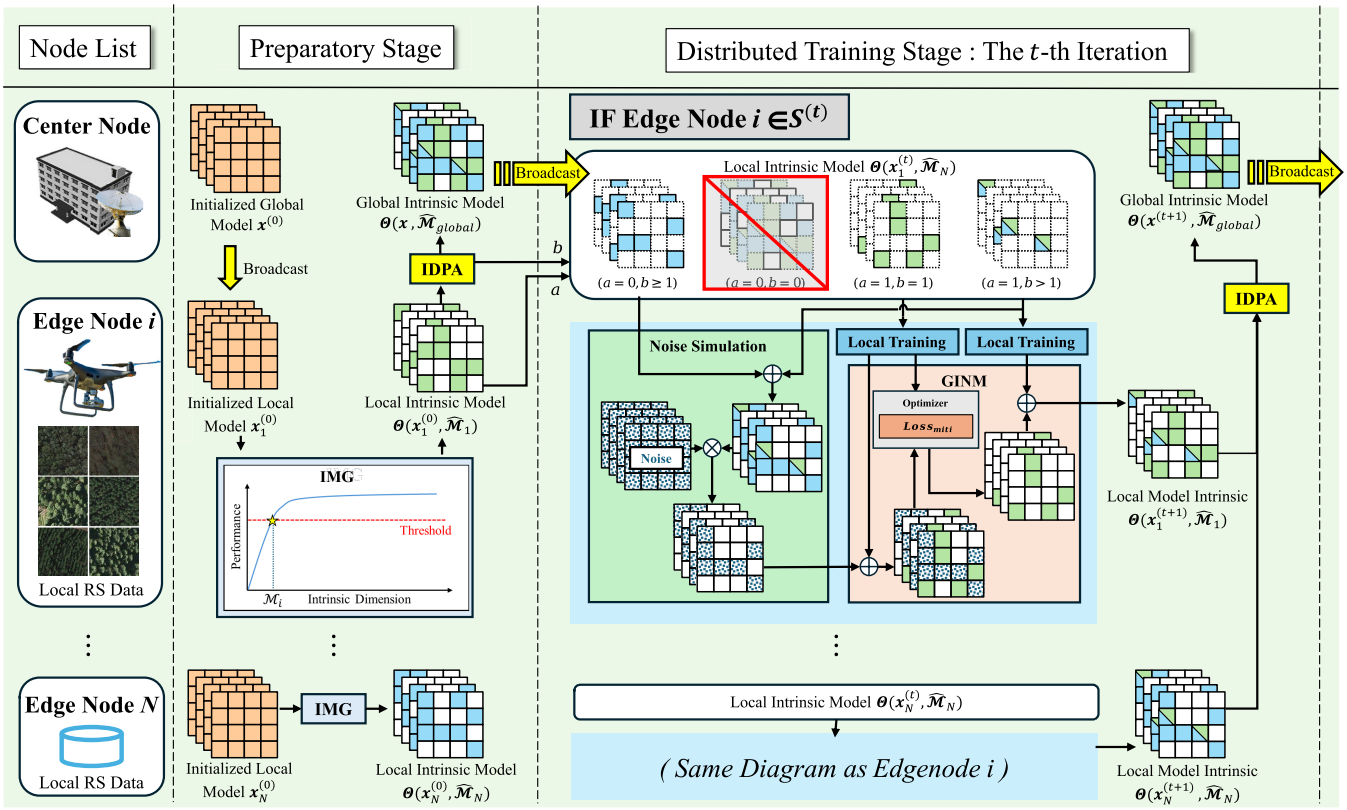


Fig. 3. In the preparatory stage, single communication is employed to generate both local and global intrinsic masks. In distributed training stage, participating nodes receive the latest global intrinsic model to initialize their local intrinsic model and perform local training. Then, nodes mitigate the impact of global intrinsic noise in the unique dimension of local intrinsic models. Finally, local intrinsic models are uploaded to the center node to update the global intrinsic model.

Algorithm 2 IMG

Require: local data ξ_i, initialized global model $\boldsymbol{x}^{(0)}$, initialized parameter masks $\boldsymbol{\mathcal{M}_i}$, epoch E hyperparameters $Acc_{\text{threshold}}$.

for $k = 1, 2, \ldots, 20$ **do**
  Train $\boldsymbol{\Theta}(\boldsymbol{x}^{(0)}, \boldsymbol{\mathcal{M}_i})$ with ξ_i^{train} for E epochs.
  $Acc \leftarrow$ evaluate $\boldsymbol{\Theta}(\boldsymbol{x}^{(0)}, \boldsymbol{\mathcal{M}_i})$ with local validation dataξ_i^{val}.
  if $Acc < Acc_{\text{threshold}}$ **then**
    $\boldsymbol{\mathcal{M}_i} \leftarrow$ Reduce sparsity:Randomly change 5% of entries from 0 to 1.
  else
    return $\boldsymbol{\mathcal{M}_i}$ as $\hat{\boldsymbol{\mathcal{M}}}_i$.
  end if
end for
return $\boldsymbol{\mathcal{M}_i}$ as $\hat{\boldsymbol{\mathcal{M}}}_i$.

Consequently, the parameter space associated with this subnetwork can be considered as a stochastic subspace within the global model's parameter space. Each node uses local data to conduct a small amount of training on the subnetwork to adjust the mask, so that the parameter subspace dimension is approximately the ID of the local model, as shown in Algorithm 2. The resulting mask is called the local intrinsic mask denoted as $\hat{\mathcal{M}}_{\mathbf{i}} = \{\hat{m}_{i,j}\}_{j=1}^{d}$, and the corresponding model is called the local intrinsic model. $\hat{\boldsymbol{x}}_i = \{\hat{x}_{i,j}\}_{j=1}^{d}$ denotes the parameters of the local intrinsic model of node i. The ID d is the effective number of parameters that truly contribute to the model's performance. The intrinsic mask $\mathcal{M}$ is a binary vector such that $\Theta(x, \mathcal{M})$ retains the parameters corresponding to the IDs, where $\mathcal{M}$ is a sparse mask with most elements being zero. The ID d is estimated through a search process during training, where the mask is iteratively adjusted to reach the threshold while maintaining efficiency by removing redundant parameters. The mask M ensures that only the critical dimensions (i.e., parameters corresponding to d) are retained, and the rest are effectively "masked out" (set to zero). Due to the massive data distribution in real scenarios, the global model is highly over-parameterized for local data, necessitating only a few local training iterations to achieve mask adjustment. Given that the local intrinsic masks are binary, sparse, and time-invariant, the generation of intrinsic masks does not impose excessive computational or storage burden compared to the conventional framework.

B. ID Parameter Averaging

The local intrinsic masks exhibit randomness, leading to distinct parameter spaces corresponding to different local subnetworks. Under such circumstances, it is not feasible to directly average parameters of local models following the conventional framework. The general approach is zero-padding the subnetwork to restore it to the full network, followed by averaging the parameters in the parameter space of the full

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

network [22], [23]. This approach exhibits two drawbacks: First, it reinstates the subnetwork to the full network, thereby compromising the model efficiency of local intrinsic models. Second, it fails to consider the mutual adversarial impact between local subnetworks of different nodes. To address these limitations, we propose a novel aggregation method called IDPA.

All local intrinsic masks are superimposed to obtain a global intrinsic mask $\hat{\boldsymbol{\mathcal{M}}}_{\text{global}}$

$$\hat{\boldsymbol{\mathcal{M}}}_{\text{global}} = \sum_{i=1}^{N} \hat{\boldsymbol{\mathcal{M}}}_i. \tag{3}$$

The global intrinsic mask projects the full network onto a new subnetwork $\boldsymbol{\Theta}(\boldsymbol{x}, \hat{\boldsymbol{\mathcal{M}}}_{\text{global}})$, resulting in the global intrinsic model with parameter $\hat{\boldsymbol{x}}_{\text{global}}$. The parameter space of this model is the union of all local intrinsic models' parameter spaces. In this parameter space, the local intrinsic models are aggregated in accordance with the following equation:

$$\hat{x}_{\text{global},j} = \sum_{i=1}^{N} \rho_j \hat{x}_{i,j} = \sum_{i=1}^{N} \frac{\hat{m}_{i,j}\hat{x}_{i,j}}{\hat{m}_{\text{global},j}}. \tag{4}$$

The efficiency of the local intrinsic model can be maximally transferred to the global intrinsic model, as the global intrinsic mask is derived from the superimposition of sparse local intrinsic masks. The participation of only the local model is permitted in the aggregation within its ID due to ρ_j in (4), thereby shielding the interference of its non-ID on other local models. In addition, only the local intrinsic models need to be transmitted between the nodes during aggregation, which can significantly reduce communication costs. IDPA effectively reduces the adversarial interference among heterogeneous local subnetworks by enforcing aggregation only within shared IDs, as defined by the superimposed global intrinsic mask. Specifically, each client contributes updates only to the parameter dimensions it actively trained, and these contributions are scaled by the participation weight ρ_j, which reflects the number of clients sharing dimension j. This avoids updating parameters from clients that have no knowledge or gradient signal on those dimensions, thereby suppressing noise propagation from mismatched subnetworks. As a result, IDPA ensures that parameter updates are concentrated in globally consistent subspaces, which mitigates destructive interference during global aggregation and promotes stable convergence.

C. Self-Distillation With Noise Simulation: Mitigating Global Intrinsic Noise

In this section, we present our approach of using self-distillation combined with noise simulation to mitigate global intrinsic noise in FL. Our method categorizes model parameters into four groups using local and global masks. Noise is applied only to the nonlocal IDs and shared IDs. During the distillation process, we fine-tune only the unique IDs of the student model, ensuring that the optimization focuses on parameters that are intrinsic to each node. Global intrinsic noise (GID) arises due to the interference between the local models in FL, especially when data across nodes are non-IID. This noise is caused by discrepancies between the updates of the local models and the global model. We define global intrinsic noise as the difference between local models and the global model

$$\sigma_{\text{GID},i} \overset{\text{def}}{=} \mathbb{E} \sum_{\psi} \left\| \text{output}\left(\hat{x}_{\text{global},j}, \xi_i\right) - \text{output}\left(\hat{x}_{i,j}, \xi_i\right) \right\| \tag{5}$$

where $\psi = \{j | \hat{m}_{i,j} = 0\} \cap \{j |\, \hat{m}_{\text{global},j} \neq 0\}$. This global intrinsic noise $\sigma_{\text{GID},i}$ measures the divergence between the local models and the global model due to non-IID data distribution. Our objective is to reduce this noise by fine-tuning only the unique IDs of the student model during self-distillation, while noise is added to the other dimensions. We classify model parameters into four groups using local and global masks. These masks help identify which parameters are intrinsic to each node and which are shared across nodes. To facilitate effective expression,

$$(a,\ b) \overset{\text{def}}{=} \hat{x}_{i,j},\ \ j \in \left\{ j \middle|\ \hat{m}_{i,j} = a \right\} \cap \left\{ j \middle|\ \hat{m}_{\text{global},j} = b \right\}. \tag{6}$$

Local IDs (LID, $(a = 1, b = 1)$): These are the parameters that are specific to the node and should only be updated locally. They correspond to the unique features of the node's local data.

Shared IDs (SID, $(a = 1, b>1)$): These are the parameters that represent features shared across nodes. They are updated during local training and parameter averaging. Therefore, the optimization of such parameters depends on multiple nodes.

Nonlocal IDs (NLID, $(a = 0, b \geq 1)$): These are the parameters that are influenced by nonlocal data, representing model features that are influenced by neighboring nodes or distant data sources. They are only updated during parameter averaging. Therefore, the optimization of such parameters is independent of node i and are the main part subject to global intrinsic noise.

Non-IDs (NID, $(a = 0, b = 0)$): Such parameters do not belong to the local intrinsic model of any node. They are not updated at all. Therefore, in consideration of the model's efficiency, such parameters are excluded from the global model.

According to the source of GID, noise simulation is applied only to NLID and SID to simulate the global model's behavior. This simulation helps the student model generalize across nodes without overfitting to local data. We ensure that the student model learns from both local and global features while minimizing the global intrinsic noise.

To simulate the global model, we add noise to the student model, particularly affecting the NLID and SID. This is done to ensure that the student model adapts to the global model while preserving its unique local features. Let the model parameters of the student model be

$$x_{\text{student}} = x_{\text{teacher}} + (\delta)_i \tag{7}$$

where$\delta_i \sim N(0, \sigma^2)$ is the Gaussian noise added to the student model's parameters. The noise is applied only to the NLID and SID of the model, as these are the parameters that need to be aligned with the global model. By perturbing these parameters, the student model is encouraged to generalize better, reducing local data bias and interference from non-IID data.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

During the distillation process, we focus on fine-tuning only the LID of the student model. These are the parameters that are intrinsic to the local data at each node. The student model is trained by minimizing the distillation loss, which ensures that it learns to replicate the global model's behavior while optimizing its unique IDs. The distillation loss function is defined as

$$\begin{aligned}\text{Loss}_{\text{miti}} = \alpha KL\Big(\text{output}\Big(\hat{\boldsymbol{x}}_i', \xi_i\Big)||\text{output}\big(\hat{\boldsymbol{x}}_i, \xi_i\big)\Big) \\ + (1-\alpha)CE\big(\xi_i, \text{output}\big(\hat{\boldsymbol{x}}_i, \xi_i\big)\big)\end{aligned} \tag{8}$$

where $\hat{\boldsymbol{x}}_i'$ denotes the parameters after adding noise. KL means the Kullback–Leibler divergence, and CE stands for cross-entropy. Both of them are measures that quantify the difference between two probability distributions. α is a the hyperparameter that controls the degree of noise mitigation. Due to the asymmetry of KL, it is more beneficial to improve the anti-interference ability of the local intrinsic model. The first item in $\text{Loss}_{\text{mitigation}}$ is the main force for noise mitigation, while the second item is for maintenance of local precision performance.

Without noise, the error between the global and student models is

$$(\epsilon)_i = \big\|\text{output}\big(x_{\text{teacher}}, (\xi)_i\big) - \text{output}\big(x_i, (\xi)_i\big)\big\|. \tag{9}$$

After adding noise to the student model, the error becomes

$$\epsilon_i' = \big\|\text{output}\big(x_{\text{teacher}}, (\xi)_i\big) - \text{output}\big(x_i + (\delta)_i, (\xi)_i\big)\big\|. \tag{10}$$

Using Taylor expansion, the new error can be approximated as

$$\epsilon_i' \approx (\epsilon)_i + \nabla\text{output}\big(x_i, (\xi)_i\big) \cdot (\delta)_i \tag{11}$$

where $\nabla\text{output}(x_i, (\xi)_i)$ is the gradient of the model output with respect to its parameters. The addition of noise decreases the discrepancy between the teacher and student models by encouraging the student model to better match the teacher model's output, even in the presence of local data perturbations. This is because noise helps align the student model with the global model, which reduces the discrepancy between local and global model updates. By adding noise only to the nonlocal IDs and shared IDs, we help the student model generalize to the global model, suppressing global intrinsic noise. During distillation, only the unique IDs of the student model are fine-tuned, which ensures that the model focuses on the relevant features and generalizes better across nodes. This process stabilizes the global model, reduces interference from non-IID data across federated nodes, and leads to a more consistent and accurate global model.

D. Discussion

In terms of scalability, RS-IML is compatible with standard gradient-based optimizers. In our experiments, we employ SGD to ensure fair comparison across baselines. However, RS-IML can be readily combined with adaptive optimizers such as Adam, and even federated-aware optimizers like FedProx [31], which incorporate gradient correction or regularization to address data heterogeneity. Using such optimizers could further enhance training stability or convergence in non-IID settings. Although not the focus of this study, exploring optimizer variants within RS-IML offers an interesting direction for future research, especially when extending to large-scale or heterogeneous model architectures.

In terms of cost, RS-IML imposes only lightweight computational and communication burdens on each client during FL. Since each client updates only a sparse intrinsic subnetwork defined by its local binary mask, the number of active parameters is significantly reduced, leading to lower FLOPs and memory consumption during local training. In terms of communication, clients transmit only the nonzero intrinsic parameters rather than full model updates which substantially reduces the uplink payload. The server aggregates only the shared IDs, avoiding unnecessary overhead from nonoverlapping parameters. These properties make RS-IML highly efficient and scalable for real-world applications, particularly in low-bandwidth or edge-device FL scenarios.

IV. Experiments

We evaluate RS-IML on two classical downstream tasks in the RS field, including RS image classification and multimodal scene classification. First, the datasets are presented in Section IV-A. Next, in Section IV-B, the evaluation metrics and experimental settings are described. In Section IV-C, the main results are described in detail and analyzed. Sections IV-D and IV-E, respectively, report the insensitivity of RS-IML to α and $Acc_{\text{threshold}}$.

A. Datasets

1) UC Merced Land-Use Dataset: The UC Merced land-use dataset (UCML-21) [74] is released in October 2010 by the Computer Vision Laboratory at the University of California, Merced. This dataset comprises images manually extracted from the urban area imagery collection of the U.S. Geological Survey (USGS). Each image has a size of 256 × 256 pixels and a spatial resolution of 1 ft. The dataset encompasses 21 categories, each containing 100 images. These categories include agricultural areas, aircraft, baseball fields, and others. Fig. 4 illustrates sample images randomly selected from the UCML-21 dataset.

2) NWPU-RESISC45 Dataset: The NWPU-RESISC45 RS image dataset (NWPU-45) [75], developed by Northwestern Polytechnical University, is a comprehensive public resource designed for the classification of RS scenes. The dataset originates from Google Earth imagery and spans more than 100 countries and regions globally. It includes 45 distinct scene categories, with each category containing 700 images. Each image has a resolution of 256 × 256 pixels. Representative samples from each category are illustrated in Fig. 5. In comparison to UCML-21, NWPU-45 stands out for its extensive scale and the rich detail present in the images.

3) HSI-LiDAR Houston2013 Dataset: The HSI-LiDAR Houston2013 dataset serves as a critical Earth observation resource that integrates hyperspectral and LiDAR data to support in-depth analysis of land cover and environmental changes. This dataset is developed by the CosmiQ Works team and incorporates high-resolution imagery from the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.



Fig. 4. Example images from the UCML-21 dataset, which includes 21 land-use scene categories: agricultural areas, airplanes, baseball diamonds, beaches, buildings, chaparral regions, dense residential zones, forests, freeways, golf courses, harbors, intersections, medium-density residential areas, mobile home parks, overpasses, parking lots, rivers, runways, sparse residential zones, storage tanks, and tennis courts.

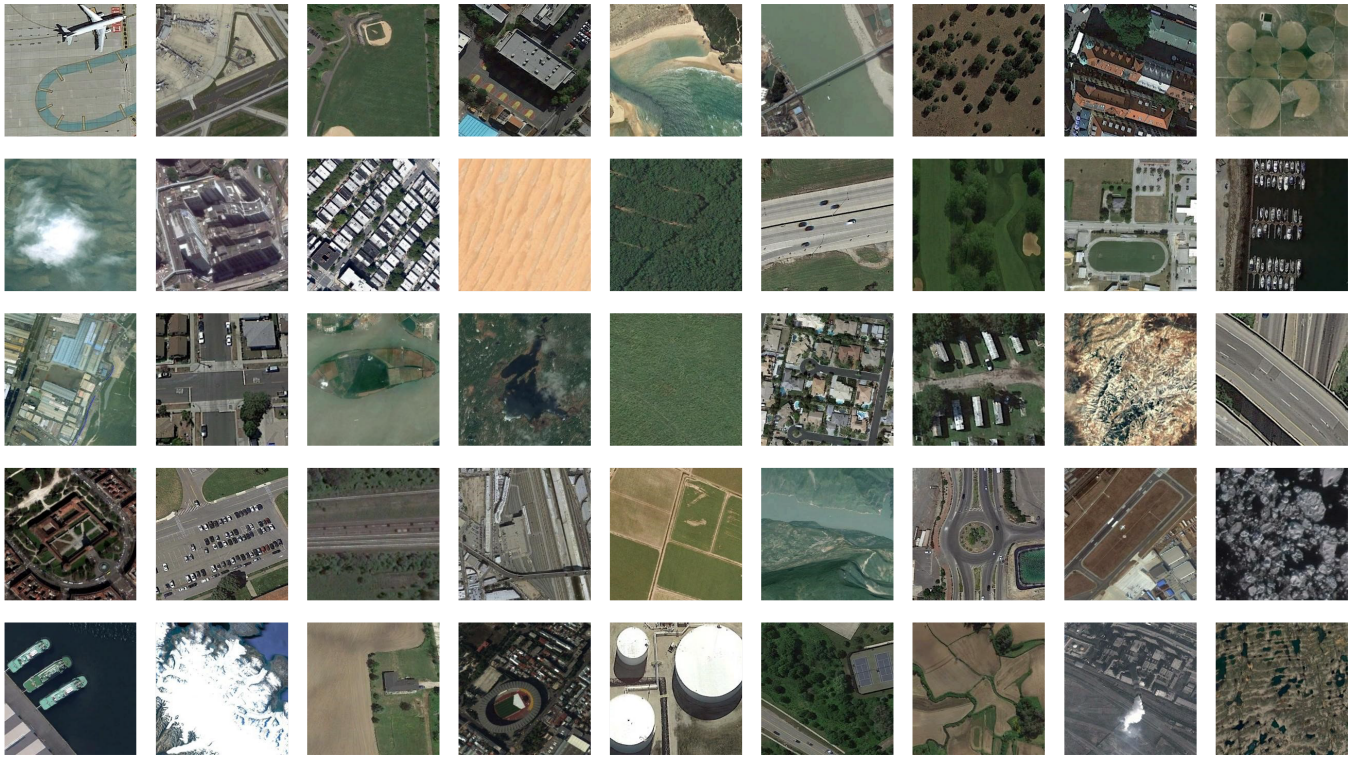


Fig. 5. Example images from the NWPU-45 dataset, which encompasses 45 land-use scene categories: airplanes, airports, baseball diamonds, basketball courts, beaches, bridges, chaparral regions, churches, circular farmlands, clouds, commercial zones, dense residential areas, deserts, forests, freeways, golf courses, ground track fields, harbors, industrial zones, intersections, islands, lakes, meadows, medium-density residential areas, mobile home parks, mountains, overpasses, palaces, parking lots, railways, railway stations, rectangular farmlands, rivers, roundabouts, runways, sea ice, ships, snowbergs, sparse residential areas, stadiums, storage tanks, tennis courts, terraces, thermal power stations, and wetlands.

WorldView-2 satellite. In 2013, an extensive data collection campaign is conducted in the University of Houston campus area, Texas, USA, utilizing the ITRES CASI1500 imaging sensor. The hyperspectral component of the dataset comprises 144 spectral bands, spanning a wavelength range of 0.38–1.05 m with a spectral interval of 10 nm. The LiDAR data consist of a single-band image. The size of all images in the dataset is 349 $\times$ 1905 pixels, as shown in Fig. 6.

B. Metrics and Setting

1) Evaluation Metrics: In RS image classification tasks, network performance is evaluated using top-1 accuracy. For

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

the multimodal scene classification task, we utilize three key metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient. OA measures the ratio of correctly classified test samples to the total number of test samples. AA reflects the mean accuracy across all categories. The kappa coefficient evaluates the agreement between the classification map produced by the model and the ground truth labels. We still employed a series of indicators related to model efficiency, including: floating point operations (FLOPs), the peak memory consumption during the local training process (PMU), and the parameter quantity of the model (*Para.*).

2) Experimental Setting: We assume a total pool of 100 nodes and that 10% of nodes participate in each round of communication ($C = 0.1$). Experiments on UCML-21 and NWPU-45 are conducted separately under IID and "pathologically Non-IID" [27] data distributions. We chose the classic FedAvg as the baseline for our comparison. We select FedDST [22], FedTiny [20], FedLP [76], and DWNP [77] as comparison methods. All experiments are conducted on an NVIDIA GTX 3090 GPU. To ensure the reliability and validity of the experiments, the models in training are implemented with Pytorch [78]. To mitigate the impact of random seeding, all results are averaged over multiple trials. When it is not necessary to specify the parameters, the local batch size is set to 50 ($B = 50$). For the RS image datasets UCML-21 and NWPU-45, the network is trained for 500 communication rounds ($R = 500$). In the multimodal scene classification task, we set the communication rounds to 30 ($R = 30$). All experiments employ the warm-up [14] algorithm to enhance convergence, specifically, the learning rate is linearly increased to 0.01 in the first three epochs and then decreases to 10^{-6} according to the cosine schedule. Other parameters are set as follows: $E = [5, 10, 5]$, $\alpha = 0.5$, $\text{Acc}_{\text{threshold}} = 50$ for IID data distributions and $E = [5, 10, 5]$, $\alpha = 0.5$, $\text{Acc}_{\text{threshold}} = 25$ for Non-IID. To ensure that the total epochs are consistent, the local training epoch of the comparison algorithm is set to 15 (Sum($E_{\text{Local_train}}$, $E_{\text{Mitigation}}$). As described in their literature, the unique parameters of the comparison algorithm are precisely configured for optimal performance.

TABLE I
NUMERICAL RESULTS ON UCML-21 UNDER IID DATA DISTRIBUTION

MODELS	METHOD	TOP-1 ACC(%)	FLOPS	PMU	PARA.
VGG-9	FEDAVG [28]	80.71	$1 \times (6.7G)$	$1 \times (1.163G)$	$1 \times (124.13e6)$
	FEDTINY [21]	71.24	0.318×	0.369×	0.3×
	FEDDST [23]	76.54	0.367×	1×	0.3×
	FEDLP [77]	75.69	0.359×	1×	0.3×
	DWNP [78]	73.24	0.334×	1×	0.3×
	RS-IML	77.63	0.323×	0.344×	0.3×
RESNET-56	FEDAVG [28]	80.48	$1 \times (6.2G)$	$1 \times (0.797G)$	$1 \times (961.25e3)$
	FEDTINY [21]	70.95	0.307×	0.354×	0.3×
	FEDDST [23]	77.68	0.376×	1×	0.3×
	FEDLP [77]	74.23	0.348×	1×	0.3×
	DWNP [78]	76.41	0.346×	1×	0.3×
	RS-IML	78.46	0.327×	0.361×	0.3×
MOBILEVIT	FEDAVG [28]	82.19	$1 \times (0.7G)$	$1 \times (0.859G)$	$1 \times (2.31e6)$
	FEDTINY [21]	59.64	0.611×	0.634×	0.6×
	FEDDST [23]	69.45	0.643×	1×	0.6×
	FEDLP [77]	68.12	0.628×	1×	0.6×
	DWNP [78]	67.23	0.633×	1×	0.6×
	RS-IML	73.34	0.625×	0.631×	0.6×

TABLE II
NUMERICAL RESULTS ON UCML-21 UNDER PATHOLOGICALLY NON-IID DATA DISTRIBUTION

MODELS	METHOD	TOP-1 ACC(%)	FLOPS	PMU	PARA.
VGG-9	FEDAVG [28]	68.93	$1 \times (6.7G)$	$1 \times (1.163G)$	$1 \times (124.13e6)$
	FEDTINY [21]	54.67	0.407×	0.433×	0.4×
	FEDDST [23]	58.64	0.417×	1×	0.4×
	FEDLP [77]	54.51	0.423×	1×	0.4×
	DWNP [78]	57.26	0.434×	1×	0.4×
	RS-IML	56.59	0.413×	0.424×	0.4×
RESNET-56	FEDAVG [28]	69.57	$1 \times (6.2G)$	$1 \times (0.797G)$	$1 \times (961.25e3)$
	FEDTINY [21]	62.14	0.355×	0.371×	0.35×
	FEDDST [23]	66.67	0.387×	1×	0.35×
	FEDLP [77]	61.47	0.366×	1×	0.35×
	DWNP [78]	65.21	0.374×	1×	0.35×
	RS-IML	67.88	0.354×	0.387×	0.35×
MOBILEVIT	FEDAVG [28]	70.83	$1 \times (0.7G)$	$1 \times (0.859G)$	$1 \times (2.31e6)$
	FEDTINY [21]	41.12	0.751×	0.831×	0.75×
	FEDDST [23]	50.63	0.787×	1×	0.75×
	FEDLP [77]	40.09	0.762×	1×	0.75×
	DWNP [78]	52.14	0.759×	1×	0.75×
	RS-IML	52.34	0.753×	0.822×	0.75×

C. Main Results

1) RS Image Classification: To ensure a comprehensive verification of the versatility of our proposed method, we conduct the RS image classification experiment on UCML-21 and NWPU-45 datasets, with various backbones including VGG-9 [79], ResNet-56 [80], and MobileViT [81]. VGG-9 and ResNet-56 are network architectures based on convolutional layers. The performance loss of the model is not obvious after lightweight. For the transformer-based network architecture MobileViT, the performance degradation of various model lightweight is relatively large. This may be since the self-attention mechanism layer in ViT involves interactions across all locations.

Tables I and II show the top-1 accuracy, FLOPs, PMU, and *Para.* compared with other methods in the experiment of UCML-21. When using VGG-9 and under IID data distribution, the model performance of RS-IML decreased by only 3.08% compared to the full-precision model (baseline) at similar model compression rate, which is better than 4.17% and 9.47% of FedDST and FedTiny. At the same time, RS-IML can significantly reduce the FLOPs, PMU, and *Para.* during training. FLOPs decreased to 0.323 × and PMU decreased to 0.344 × compared to baseline. In comparison, RS-IML demonstrates comparable efficiency improvements to FedTiny, yet it offers superior performance stability for the model. Given that FedDST requires the recovery of a lightweight model during aggregation, RS-IML has a clear advantage over PMU. Similar results also appeared in the ResNet56 experiment, where the performance of RS-IML model decreased by only 2.02% compared with the full-precision model, which is better than 2.8% and 9.53% of FedDST and FedTiny. When using MobileViT, the performance degradation is relatively large. However, when compared with other methods,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

TABLE III
NUMERICAL RESULTS ON NWPU-45 UNDER IID DATA DISTRIBUTION

MODELS	METHOD	TOP-1 ACC(%)	FLOPS	PMU	PARA.
VGG-9	FEDAVG [28]	76.67	$1\times(6.7G)$	$1\times(1.164G)$	$1\times(124.23e6)$
	FEDTINY [21]	69.45	0.318×	0.369×	0.3×
	FEDDST [23]	73.24	0.367×	1×	0.3×
	FEDLP [77]	74.64	0.397×	1×	0.3×
	DWNP [78]	73.01	0.341×	1×	0.3×
	RS-IML	75.12	0.322×	0.373×	0.3×
RESNET-56	FEDAVG [28]	77.84	$1\times(6.2G)$	$1\times(0.798G)$	$1\times(1.09e6)$
	FEDTINY [21]	70.34	0.307×	0.354×	0.3×
	FEDDST [23]	74.69	0.376×	1×	0.3×
	FEDLP [77]	73.24	0.369×	1×	0.3×
	DWNP [78]	71.12	0.347×	1×	0.3×
	RS-IML	75.98	0.324×	0.355×	0.3×
MOBILEVIT	FEDAVG [28]	78.23	$1\times(0.7G)$	$1\times(0.863G)$	$1\times(2.31e6)$
	FEDTINY [21]	63.11	0.611×	0.634×	0.6×
	FEDDST [23]	70.15	0.643×	1×	0.6×
	FEDLP [77]	68.24	0.684×	1×	0.6×
	DWNP [78]	66.64	0.645×	1×	0.6×
	RS-IML	72.68	0.618×	0.626×	0.6×

TABLE IV
NUMERICAL RESULTS ON NWPU-45 UNDER PATHOLOGICALLY NON-IID DATA DISTRIBUTION

MODELS	METHOD	TOP-1 ACC(%)	FLOPS	PMU	PARA.
VGG-9	FEDAVG [28]	65.65	$1\times(6.7G)$	$1\times(1.164G)$	$1\times(124.23e6)$
	FEDTINY [21]	53.12	0.407×	0.433×	0.4×
	FEDDST [23]	59.74	0.417×	1×	0.4×
	FEDLP [77]	51.62	0.432×	1×	0.4×
	DWNP [78]	57.41	0.421×	1×	0.4×
	RS-IML	60.12	0.411×	0.446×	0.4×
RESNET-56	FEDAVG [28]	68.34	$1\times(6.2G)$	$1\times(0.798G)$	$1\times(1.09e6)$
	FEDTINY [21]	59.12	0.355×	0.371×	0.35×
	FEDDST [23]	63.01	0.387×	1×	0.35×
	FEDLP [77]	53.24	0.388×	1×	0.35×
	DWNP [78]	66.94	0.367×	1×	0.35×
	RS-IML	63.13	0.353×	0.389×	0.35×
MOBILEVIT	FEDAVG [28]	67.24	$1\times(0.7G)$	$1\times(0.863G)$	$1\times(2.31e6)$
	FEDTINY [21]	34.65	0.751×	0.831×	0.75×
	FEDDST [23]	49.68	0.787×	1×	0.75×
	FEDLP [77]	34.89	0.794×	1×	0.75×
	DWNP [78]	45.66	0.756×	1×	0.75×
	RS-IML	51.14	0.752×	0.824×	0.75×

the method we proposed is still better, with top-1 accuracy reaching 52.34%, that is better than 50.63% and 41.12% of FedDST and FedTiny. When using VGG-9 and under Non-IID data distribution, top-1 accuracies is significantly reduced due to the data heterogeneity, and the harm of model lightweighting to performance will also increase, but RS-IML remains effective. A possible explanation is that the suppression of global intrinsic noise by RS-IML not only reduces the adverse effects between local models, but also preserves the components of mutual improvement between them, thus reducing the harm of model lightweight on performance. In terms of trained FLOPS and peak memory occupancy, RS-IML has a relatively ideal model efficiency because it has not trained the full-precision model in the whole process and can aggregate local eigenmodels with certain differences without parameter completion. Compared with FedLP based on layer pruning and DWNP based on channel pruning, RS-IML demonstrates a clear advantage in terms of PMU, as it eliminates the need to train dense models.

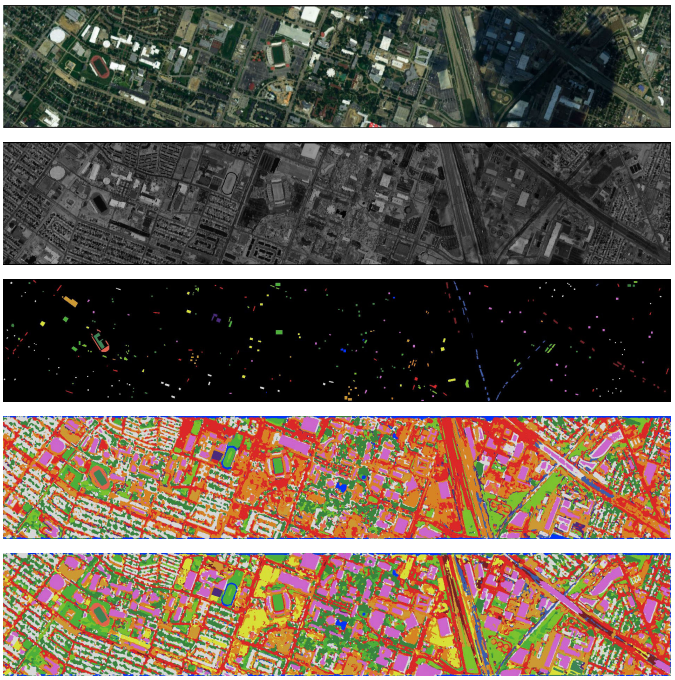


Fig. 6. Visualization of scene display and classification results for the Houston2013 dataset. The images are arranged from top to bottom as follows: the hyperspectral image (HSI) of the Houston2013 dataset, the LiDAR imagery, the ground truth labels, the scene classification outcomes from the baseline model, and the classification outcomes from the RS-IML model.

TABLE V
NUMERICAL RESULTS ON THE HSI-LIDAR HOUSTON2013 DATASET

MODELS	OA(%)	AA(%)	KAPPA	FLOPS	PARA.
FEDAVG [28]	90.87	90.67	90.43	$1\times(3.51G)$	$1\times(2.3e5)$
FEDTINY [21]	87.15	87.09	86.88	0.201×	0.2×
FEDDST [23]	89.94	89.73	89.42	0.254×	0.2×
RS-IML	90.33	90.17	90.02	0.217×	0.2×

Tables III and IV show the Top-1 accuracy, FLOPs, PMU, and *Para.* compared with other methods in the experiment of NWPU-45. RS-IML achieves the least performance degradation (1.55%) when using VGG9 under IID data distribution. At the same time, it also has certain advantages over FedDST, FedLP and DWNP in terms of FLOPs, PMU, and *Para.* in training.

2) Multimodal Scene Classification: To ensure a verification of the versatility of our proposed method, we also conducted experiments focusing on RS multimodal scene classification using an advanced multimodal fusion networks, namely ExViT [82], a transformer-based network architecture. The experimental results are shown in Table V. We also give a visualization of the classification results. As shown in Fig. 6, the proposed algorithm can still achieve good performance compared with the comparison algorithm.

D. Performance With Different α

α determines the strength of RS-IML to suppress global intrinsic noise, which will have a certain impact on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
NUMERICAL RESULTS WITH DIFFERENT α

α	0.00	0.25	0.5	0.75	1.00
TOP-1 ACC(%)	70.41	75.12	78.46	78.23	74.18

performance of the final model. The model used in the experiment is ResNet-56, and the dataset is UCML-21 with non-IID distribution. ($\text{Acc}_{\text{threshold}} = 80$, $\alpha = [0, 0.25, 0.5, 0.75, 1]$, $E = [5, 10, 5]$). In Table VI, we show that RS-IML performs well with any reasonable value of α, even in the non-IID settings we explore in this article. When $\alpha = 0$, there is no correction, and the model performs at a 70.41% top-1 accuracy, which reflects poor performance due to the absence of global noise mitigation. For $\alpha = 0.25$, the performance improves to 75.12%, indicating that a small amount of noise perturbation provides a better balance. At $\alpha = 0.50$, 78.46% accuracy is achieved, which is the optimal value for global noise suppression and model accuracy, as reflected in the results from Table VI. At $\alpha = 0.75$, the accuracy stabilizes at 78.23%, and further increasing α to 1.00 results in a slight drop in performance to 74.18%. This suggests that the model benefits from noise mitigation up to $\alpha = 0.5$, beyond which the noise becomes too strong, leading to loss of precision in the model's predictions. Thus, we find that $\alpha = 0.50$ offers the best trade-off between noise suppression and model performance. Smaller or larger values lead to either insufficient noise mitigation or over-correction, causing degradation in model performance.

E. Performance With Different $Acc_{\text{threshold}}$

$\text{Acc}_{\text{threshold}}$ is closely related to the efficiency of the final model. The higher the precision threshold, the more stringent the estimation of the ID of the local model by RS-IML. The model used in the experiment is ResNet-56, and the dataset is UCML-21 with non-IID distribution. ($\text{Acc}_{\text{threshold}} = [15, 35, 55, 75]$, $\alpha = 0.5$, $E = [5, 10, 5]$). Not surprisingly, in TableVII, we show that RS-IML's performance is related to $\text{Acc}_{\text{threshold}}$.

When $\text{Acc}_{\text{threshold}}$ is set to 15%, the model shows a significant drop in performance, with top-1 accuracy reaching only 21.45%. The model struggles to capture enough meaningful features due to an insufficient estimation of IDs. At $\text{Acc}_{\text{threshold}}$ = 35%, performance improves to 78.46% top-1 accuracy, indicating a reasonable trade-off between retaining relevant features and minimizing unnecessary parameters. At $\text{Acc}_{\text{threshold}}$ = 55%, 79.15% accuracy is achieved. Further increasing $\text{Acc}_{\text{threshold}}$ leads to diminishing returns, as the model becomes more efficient but struggles with overfitting. At $\text{Acc}_{\text{threshold}}$ = 75%, the top-1 accuracy increases slightly to 80.13%, but it requires significant computational resources (as shown in Table VII, with higher FLOPs, PMU, and parameters). From this analysis, it is evident that $\text{Acc}_{\text{threshold}}$ values in the range of 35%–55% provide the best performance while maintaining a balance between efficiency and accuracy. Very high $\text{Acc}_{\text{threshold}}$ values (e.g., 75%) may improve accuracy but at the cost of excessive computational requirements. With the increase of $\text{Acc}_{\text{threshold}}$, the performance requirements of the local intrinsic model becomes higher, which is conducive to obtaining the global intrinsic model with better performance. However, this may lead to an inclination of the node toward retaining more parameters as intrinsic parameters, thereby impeding the attainment of higher model efficiency.

TABLE VII
PERFORMANCE WITH DIFFERENT $\text{ACC}_{\text{threshold}}$

$Acc_{\text{threshold}}(\%)$	TOP-1 ACC(%)	FLOPS	PMU	PARA.
15	21.45	0.131×	0.144×	0.1×
35	78.46	0.327×	0.361×	0.3×
55	79.15	0.534×	0.561×	0.55×
75	80.13	0.936×	0.957×	0.9×

V. CONCLUSION

This paper introduces RS-IML, a novel FL framework for RS-IML without training dense models. RS-IML addresses the issues of reliance on initial training of dense models during lightweighting and potential performance degradation after lightweighting. It proposes three new components: IMG, IDPA, and global intrinsic noise mitigation. IMG lightweight neural networks without training dense models. IDPA aggregate inconsistent local intrinsic models while suppressing the adverse effects of nonintrinsic parameters between local models. Global intrinsic noise mitigation allows fine-tuning the specific parameters for better performance. Experimental results demonstrate that RS-IML outperforms existing approaches in terms of both accuracy and model efficiency during lightweighting.

REFERENCES

[1] W. Xie, Z. Wang, J. Ma, D. Li, and Y. Li, "RS-DGC: Exploring neighborhood statistics for dynamic gradient compression on remote sensing image interpretation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622112.

[2] Q. Bi, K. Qin, H. Zhang, J. Xie, Z. Li, and K. Xu, "APDC-net: Attention pooling-based convolutional network for aerial scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1603–1607, Sep. 2020.

[3] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, Feb. 2019.

[4] K. Xu, H. Huang, Y. Li, and G. Shi, "Multilayer feature fusion network for scene classification in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 11, pp. 1894–1898, Nov. 2020.

[5] Y. Gao, M. Zhang, W. Li, X. Song, X. Jiang, and Y. Ma, "Adversarial complementary learning for multisource remote sensing classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505613.

[6] M. Appel, F. Lahn, W. Buytaert, and E. Pebesma, "Open and scalable analytics of large Earth observation datasets: From scenes to multidimensional arrays using SciDB and GDAL," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 47–56, Apr. 2018.

[7] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[8] S. Veraverbeke et al., "Hyperspectral remote sensing of fire: State-of-the-art and future perspectives," *Remote Sens. Environ.*, vol. 216, pp. 105–121, Oct. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

[9] B. Denby and B. Lucia, "Orbital edge computing: Nanosatellite constellations as a new class of computer system," in *Proc. 25th Int. Conf. Architectural Support Program. Lang. Operating Syst.*, Mar. 2020, pp. 939–954.

[10] H. Lin, C. Zhang, D. Hong, K. Dong, and C. Wen, "FedRSClip: Federated learning for remote sensing scene classification using vision-language models," 2025, *arXiv:2501.02461*.

[11] D. Vasisht, J. Shenoy, and R. Chandra, "L2D2: Low latency distributed downlink for leo satellites," in *Proc. ACM Special Interest Group Data Commun. (SIGCOMM)*, 2021, pp. 151–164.

[12] M. M. Coffer, "Balancing privacy rights and the production of high-quality satellite imagery," American Chemical Society (ACS), Washington, DC, USA, Tech. Rep., 2020. [Online]. Available: https://pubs.acs.org/doi/pdf/10.1021/acs.est.0c02365?ref=article_openPDF

[13] J. B. Dean et al., "Large scale distributed deep networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Dec. 2012, pp. 1223–1231.

[14] P. Goyal et al., "Accurate, large minibatch SGD: Training ImageNet in 1 hour," 2017, *arXiv:1706.02677*.

[15] H. Doyu, R. Morabito, and J. Hoeller, "TinyML as-a-service: Bringing ML inference to the deepest IoT edge," TinyML Foundation, USA, Tech. Rep., Sep. 2020. [Online]. Available: https://cms.tinyml.org/wp-content/uploads/talks2020/tinyML_Talks_Hiroshi_Doyu_200915.pdf

[16] R. Vijay, "Enabling ultra-low power machine learning at the edge," in *Proc. TinyML Summit (TinyML)*, 2020, p. 851.

[17] B. Li et al., "Random sketch learning for deep neural networks in edge computing," *Nature Comput. Sci.*, vol. 1, no. 3, pp. 221–228, Mar. 2021.

[18] J. Park, S. Samarakoon, M. Bennis, and M. Debbah, "Wireless network intelligence at the edge," *Proc. IEEE*, vol. 107, no. 11, pp. 2204–2239, Nov. 2019.

[19] L. Zhu, H. Lin, Y. Lu, Y. Lin, and S. Han, "Delayed gradient averaging: Tolerate the communication latency for federated learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, Dec. 2021, pp. 29995–30007.

[20] H. Huang, L. Zhang, C. Sun, R. Fang, X. Yuan, and D. Wu, "Distributed pruning towards tiny neural networks in federated learning," 2022, *arXiv:2212.01977*.

[21] U. Evci, T. Gale, J. Menick, P. S. Castro, and E. Elsen, "Rigging the lottery: Making all tickets winners," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jan. 2019, pp. 2943–2952.

[22] S. Bibikar, H. Vikalo, Z. Wang, and X. Chen, "Federated dynamic sparse training: Computing less, communicating less, yet learning better," in *Proc. Assoc. Advancement Artif. Intell. (AAAI)*, vol. 36, Jun. 2022, pp. 6080–6088.

[23] A. Li et al., "LotteryFL: Personalized and communication-efficient federated learning with lottery ticket hypothesis on non-IID datasets," 2020, *arXiv:2008.03371*.

[24] A. Ansuini, A. Laio, J. H. Macke, and D. Zoccolan, "Intrinsic dimension of data representations in deep neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Jan. 2019, pp. 6111–6122.

[25] C. Li, H. Farkhoor, R. Liu, and J. Yosinski, "Measuring the intrinsic dimension of objective landscapes," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2018, pp. 1764–1787.

[26] X. Ma et al., "Dimensionality-driven learning with noisy labels," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018.

[27] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Statist. (AISTATS)*, Jan. 2016, pp. 1273–1282.

[28] E. B. P. Kairouz and H. B. Mcmahan, *Advances and Open Problems in Federated Learning*, vol. 14. Hanover, MA, USA: Now Publishers Inc., 2021.

[29] H. Wang, M. Yurochkin, Y. Sun, D. Papailiopoulos, and Y. Khazaeni, "Federated learning with matched averaging," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jan. 2020, pp. 12432–12447.

[30] J. Hong, Z. Zhu, S. Yu, Z. Wang, H. H. Dodge, and J. Zhou, "Federated adversarial debiasing for fair and transferable representations," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, Aug. 2021, pp. 617–627.

[31] T. Li, A. Kumar Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," 2018, *arXiv:1812.06127*.

[32] M. Duan et al., "FedGroup: Efficient clustered federated learning via decomposed data-driven measure," 2020, *arXiv:2010.06870*.

[33] T.-M. Harry Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," 2019, *arXiv:1909.06335*.

[34] R. Liu et al., "No one left behind: Inclusive federated learning over heterogeneous devices," 2022, *arXiv:2202.08036*.

[35] C. Wu, F. Wu, R. Liu, L. Lyu, Y. Huang, and X. Xie, "FedKD: Communication efficient federated learning via knowledge Distillation," *Nature Commun.*, vol. 13, no. 1, p. 2032, Aug. 2021.

[36] W. Li, Z. Wu, T. Chen, L. Li, and Q. Ling, "Communication-censored distributed stochastic gradient descent," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6831–6843, Nov. 2022.

[37] L. Wang, H. Zhong, R. Ranjan, A. Zomaya, and P. Liu, "Estimating the statistical characteristics of remote sensing big data in the wavelet transform domain," *IEEE Trans. Emerg. Topics Comput.*, vol. 2, no. 3, pp. 324–337, Sep. 2014.

[38] M. M. U. Rathore, A. Paul, A. Ahmad, B.-W. Chen, B. Huang, and W. Ji, "Real-time big data analytical architecture for remote sensing application," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 10, pp. 4610–4621, Oct. 2015.

[39] V. Syrris, C. Corbane, M. Pesaresi, and P. Soille, "Mosaicking Copernicus Sentinel-1 data at global scale," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 547–557, Sep. 2020.

[40] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[41] Y. Ma et al., "Remote sensing big data computing: Challenges and opportunities," *Future Gener. Comput. Syst.*, vol. 51, pp. 47–60, Oct. 2015.

[42] M. Chi, A. Plaza, J. A. Benediktsson, Z. Sun, J. Shen, and Y. Zhu, "Big data for remote sensing: Challenges and opportunities," *Proc. IEEE*, vol. 104, no. 11, pp. 2207–2219, Nov. 2016.

[43] J. Sun et al., "An efficient and scalable framework for processing remotely sensed big data in cloud computing environments," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4294–4308, Jul. 2019.

[44] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[45] M. Bi, M. Wang, Z. Li, and D. Hong, "Vision transformer with contrastive learning for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 738–749, 2023.

[46] Z. Han et al., "Multisource collaborative domain generalization for cross-scene remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535815.

[47] W. Boulila, M. Sellami, M. Driss, M. Al-Sarem, M. Safaei, and F. A. Ghaleb, "RS-DCNN: A novel distributed convolutional-neural-networks based-approach for big remote-sensing image classification," *Comput. Electron. Agricult.*, vol. 182, Mar. 2021, Art. no. 106014.

[48] D. Li, W. Xie, Y. Li, and L. Fang, "FedFusion: Manifold-driven federated learning for multi-satellite and multi-modality fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500813.

[49] X. Zhang, B. Zhang, W. Yu, and X. Kang, "Federated deep learning with prototype matching for object extraction from Very-High-Resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603316.

[50] J. Tan et al., "Bridging data islands: Geographic heterogeneity-aware federated learning for collaborative remote sensing semantic segmentation," 2024, *arXiv:2404.09292*.

[51] M. J. Khan, S. Rath, and M. H. Zaib, "Privacy-enhanced image restoration in remote sensing via federated learning," in *Proc. Int. Symp. Digit. Forensics Secur. (ISDFS)*, Apr. 2024, pp. 1–6.

[52] P. Tam, S. Math, C. Nam, and S. Kim, "Adaptive resource optimized edge federated learning in real-time image sensing classifications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10929–10940, 2021.

[53] I. Abunadi et al., "Federated learning with blockchain assisted image classification for clustered UAV networks," in *Proc. Comput. Mater. Contin*, vol. 72, Jan. 2022, pp. 1195–1212.

[54] J. Zhu, J. Wu, A. K. Bashir, Q. Pan, and W. Yang, "Privacy-preserving federated learning of remote sensing image classification with dishonest majority," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4685–4698, 2023.

[55] J. H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic encryption for arithmetic of approximate numbers," in *Proc. 23rd Int. Conf. Theory Appl. Cryptol. Inf. Secur.*, Jan. 2017, pp. 409–437.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.

[56] B. Büyüktaş, G. Sumbul, and B. Demir, "Learning across decentralized multi-modal remote sensing archives with federated learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 4966–4969.

[57] A.-K. Duong, H.-Â. Lê, and M.-T. Pham, "Leveraging feature communication in federated learning for remote sensing image classification," 2024, *arXiv:2403.13575*.

[58] D. Li, W. Xie, Z. Wang, Y. Lu, Y. Li, and L. Fang, "FedDiff: Diffusion model driven federated learning for multi-modal and multi-clients," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 10353–10367, Oct. 2024.

[59] L. Janson, W. Fithian, and T. J. Hastie, "Effective degrees of freedom: A flawed metaphor," *Biometrika*, vol. 102, no. 2, pp. 479–485, Jun. 2015.

[60] T. Gao and V. Jojic, "Degrees of freedom in deep neural networks," 2016, *arXiv:1603.09260*.

[61] L. Blier and Y. Ollivier, "The description length of deep learning models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 31, Jan. 2018, pp. 2216–2226.

[62] R. Ghrist, "Configuration spaces and braid groups on graphs in robotics," 2009, *arXiv: math/9905023*.

[63] W. Zhu, Q. Qiu, J. Huang, R. Calderbank, G. Sapiro, and I. Daubechies, "LDMNet: Low dimensional manifold regularized neural networks," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 2743–2751.

[64] B. Cheung, A. V. Terekhov, Y. Chen, P. Agrawal, and B. A. Olshausen, "Superposition of many models into one," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, Feb. 2019, pp. 10867–10876.

[65] T. Birdal, A. Lou, L. Guibas, and U. Simsekli, "Intrinsic dimension, persistent homology and generalization in neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Jan. 2021, pp. 6776–6789.

[66] F. Gressmann, Z. Eaton-Rosen, and C. Luschi, "Improving neural network training in low dimensional random bases," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Jan. 2020, pp. 12140–12150.

[67] S. Fort, G. K. Dziugaite, M. Paul, S. Kharaghani, D. M. Roy, and S. Ganguli, "Deep learning versus kernel learning: An empirical study of loss landscape geometry and the time evolution of the neural tangent kernel," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, Oct. 2020, pp. 5850–5861.

[68] R. K. Mahabadi, J. Henderson, and S. Ruder, "Compacter: Efficient low-rank hypercomplex adapter layers," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Jan. 2021, pp. 1022–1035.

[69] A. Panahi, S. Saeedi, and T. Arodz, "Shapeshifter: A parameter-efficient transformer using factorized reshaped matrices," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, Dec. 2021, pp. 1337–1350.

[70] J. Huang, P. Majumder, S. Kim, A. Muzahid, K. H. Yum, and E. J. Kim, "Communication algorithm-architecture co-design for distributed deep learning," in *Proc. Int. Symp. Comput. Archit. (ISCA)*, Jun. 2021, pp. 181–194.

[71] T. Sun, D. Li, and B. Wang, "Decentralized federated averaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4289–4301, Apr. 2023.

[72] Y. Wei, Y. Zhang, Q. Song, X. Zhou, Y. Zhou, and Y. Shen, "Effects of different configurations and gradients on compression responses of gradient honeycombs via selective laser melting," *Thin-Walled Struct.*, vol. 170, Jan. 2022, Art. no. 108462.

[73] L. Abrahamyan, Y. Chen, G. Bekoulis, and N. Deligiannis, "Learned gradient compression for distributed deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7330–7344, Dec. 2022.

[74] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[75] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[76] Z. Zhu et al., "FedLP: Layer-wise pruning mechanism for communication-computation efficient federated learning," in *Proc. Int. Conf. Commun. (ICC)*, Jan. 2023, pp. 1250–1255.

[77] S. Gao, J. Li, Z. Zhang, Y. Zhang, W. Cai, and H. Huang, "Device-wise federated network pruning," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 7, Jun. 2024, pp. 12342–12352.

[78] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2019, pp. 8026–8037.

[79] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[80] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[81] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," 2021, *arXiv:2110.02178*.

[82] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.

[83] Y. Dandi, L. Barba, and M. Jaggi, "Implicit gradient alignment in distributed and federated learning," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Jun. 2022, vol. 36, no. 6, pp. 6454–6462.

[84] Y. H. Ezzeldin, S. Yan, C. He, E. Ferrara, and A. S. Avestimehr, "FairFed: Enabling group fairness in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, 2023, pp. 7494–7502.

[85] L. Li, W. Xu, T. Chen, G. B. Giannakis, and Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets," in *Proc. Conf. Artif. Intell. (AAAI)*, 2019, pp. 1544–1551.

[86] F. Liu, X. Wu, S. Ge, W. Fan, and Y. Zou, "Federated learning for vision-and-language grounding problems," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 7, 2020, pp. 11572–11579.

[87] M. Liu, "Algorithmic foundation of federated learning with sequential data," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Mar. 2024, vol. 38, no. 20, p. 22675.

[88] L. Nagalapatti and R. Narayanam, "Game of gradients: Mitigating irrelevant clients in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 10, 2021, pp. 9046–9054.

[89] S. Shen, Y. Cheng, J. Liu, and L. Xu, "STL-SGD: Speeding up local SGD with stagewise communication period," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, pp. 9576–9584.

[90] V. S. Sheng and J. Zhang, "Machine learning with crowdsourcing: A brief summary of the past research and future directions," in *Proc. AAAI Conf. Artif. Intell.*, Jul. 2019, pp. 9837–9843.

[91] H. Ye, W. Xie, Y. Li, and L. Fang, "AdaGK-SGD: Adaptive global knowledge guided distributed stochastic gradient descent," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Apr. 2025, vol. 39, no. 21, pp. 22119–22127.



Hangyu Ye received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2021, where he is currently pursuing the Ph.D. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

His research interests include federated learning, distributed machine learning, image classification, object detection, and change detection.



Weiying Xie (Senior Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2017.

She is currently a Professor with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 50 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTION ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON CYBERNETICS, Conference on IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR), and Association for the Advancement of Artificial Intelligence (AAAI). Her research interests include neural networks, machine learning, hyperspectral image processing, and high-performance computing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.



Xin Zhang received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2023, where she is currently pursuing the Ph.D. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

Her research interests include model compression and inference acceleration.



Yibing Lu received the B.E. degree in software engineering from Xidian University, Xi'an, China, in 2023, where she is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

Her research interests include multimodal image processing, remote sensing classification, and target detection.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is a Professor. He is also the Director of the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center. His research interests include image and video processing, hyperspectral image (HSI) processing, and high-performance computing.



Leyuan Fang (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University, Durham, NC, USA. He is currently a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang was a recipient of the One Second-Grade National Award at the Nature and Science Progress of China in 2019. He is an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING, *Transaction on Neural Networks and Learning Systems*, and *Neurocomputing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:36:26 UTC from IEEE Xplore. Restrictions apply.